AP CALCULUS AB

Unit 1 • Lesson 1-6 • Notes (filled in)

Lesson 1-6 • Average Rate of Change

Notes (filled in)

Learning goals

- Compute an average rate of change from a formula, a table or a graph.
- Interpret an average rate of change in context with units, including average velocity.
- Use the difference quotient $\frac{f(a+h)-f(a)}{h}$.

Vocabulary. average rate of change, secant line, difference quotient, average velocity, units.

Key idea

The average rate of change of f on $[a,b]$ is $\frac{f(b)-f(a)}{b-a}$, the slope of the secant line through $(a,f(a))$ and $(b,f(b))$. On $[a,a+h]$ it is the **difference quotient** $\frac{f(a+h)-f(a)}{h}$.

Example 1 • Formula

Find the average rate of change of $f(x)=x^3-2x$ on $[-1,2]$, and simplify $\frac{f(1+h)-f(1)}{h}$ for $f(x)=x^2$.

$\frac{4-1}{3}=1$. $\frac{(1+h)^2-1}{h}=\frac{2h+h^2}{h}=2+h$.

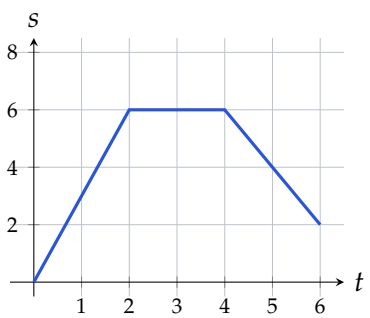
Example 2 • Table, with units

t (min)	0	5	10	15	20
$A(t)$ (L)	40	52	61	66	68

Find and interpret the average rate of change of A on $[5,15]$.

$\frac{66-52}{10}=1.4$. From $t=5$ to $t=15$ minutes, the amount of water increased on average by 1.4 liters per minute.

Example 3 • Average velocity from a graph



The position $s(t)$ (meters) of a particle is graphed. Find its average velocity on $[0,2]$, $[2,4]$ and $[0,6]$.

$\frac{6-0}{2} = 3$ m/s. $\quad 0$ m/s. $\quad \frac{2-0}{6} = \frac{1}{3}$ m/s.

Common mistakes

- Giving the change $f(b) - f(a)$ instead of the rate.
- Forgetting units, or interpreting an average rate as a rate at an instant.

Check your understanding

1. The average rate of change of $g(x) = \frac{12}{x}$ on $[2, 6]$ is $__-1__$.
2. For $f(x) = x^2 + x$, $\frac{f(2+h)-f(2)}{h} = __5+h__$.